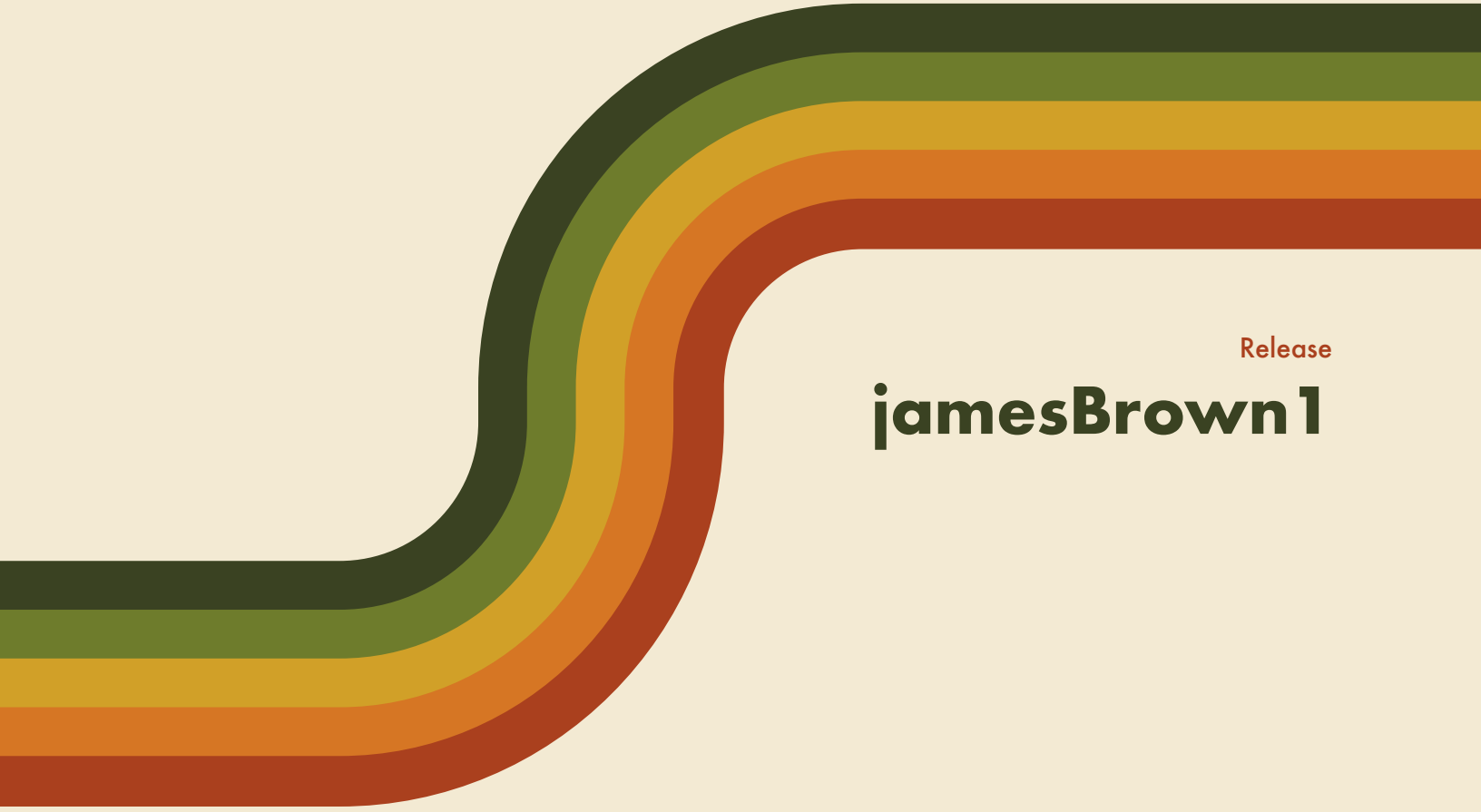
TFR

Installation Guide

Booting the TFR/911H from a FujiNet



Release

jamesBrown1

Order No. TFR-911H-IG
Release jamesBrown1, October 2026

the tfr/9 project

TFR Installation Guide

Booting the TFR/911H from a FujiNet: Release jamesBrown1

This guide tells how to install release jamesBrown1 of the TFR/911H software: how to flash the FujiNet's firmware with FujiNet-Flasher, make its SD card, flash the TFR/911H's firmware, install the tether, wire the two boards together, and boot NitrOS-9 Level 2 from the FujiNet.

Order Number:	TFR-911H-IG
Release:	jamesBrown1, 20 October 2026
Revision/Update Information:	This is a new guide.
Operating System and Version:	NitrOS-9/6809 Level 2 V3.3.0
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Hardware:	TFR/911H; an ESP32 DevKitC with 8 MB of flash and PSRAM

TFR Installation Guide, Release jamesBrown1, 20 October 2026

Revision history

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About This Guide

This guide takes you from a bare TFR/911H and an ESP32 board to NitrOS-9 Level 2 booting from the FujiNet's SD card, with the network programs working: the weather, the news, stock quotes, Bounce World, and Fuji Battleship. Every file it asks for is in the release, jamesBrown1.

Audience

This guide is for anyone setting up a TFR/911H with a FujiNet: no knowledge of OS-9, the RP2350B, or the ESP32 is assumed. You need to be able to copy files, open a terminal window and type commands in it, and push three jumper wires onto header pins.

Organization

This guide has seven chapters and three appendices:

Chapter 1 What you are building, what is in the release, what you need, and the installation at a glance.
Chapter 2 Flashing the FujiNet's firmware with FujiNet-Flasher.
Chapter 3 Making the FujiNet's SD card: the boot disk and the FujiNet's settings.
Chapter 4 Flashing the TFR/911H's firmware, and installing the tether.
Chapter 5 Wiring the FujiNet to the TFR/911H, and checking the link.
Chapter 6 Booting NitrOS-9 from the FujiNet, and what to do next.
Chapter 7 Troubleshooting.
Appendix A The release: its files, their checksums, their sources, and how it was tested.
Appendix B Building the release from source.
Appendix C Pins and settings, for reference.

Related Documents

- The *TFR Manual and Reference* (TFR-911H-MR), in the release: NitrOS-9 on the TFR/911H, its commands, the FujiNet, and how it all works.
- `v4/nitros9-l2/fujinet/README.md` in the TFR/9 source, on setting up and programming the FujiNet.
- The FujiNet project's pages, at `fujinet.online`, and its wiki, at `github.com/FujiNetWIFI/fujinet-firmware/wiki`.

Conventions

Convention	Meaning
`$`	A shell prompt on your computer.
`TCL>`	The TFR/911H firmware's prompt, before an OS boots.
`OS9:`	The NitrOS-9 Shell prompt.
`bold type`	In examples, what you type.
italic	Something you supply: a file name, a device.
Ctrl/]	Hold down the Ctrl key while you press].
FLASH	A button on a board, by the name printed beside it.
J3-9	Pin 9 of the header marked J3.
IO33	A pin of the ESP32 board, by the name printed beside it: here, GPIO 33.
`jamesBrown1/`	A file or folder of the release.

Before You Start 1

This chapter describes what you are building, what is in the release, and what you need, and it lists the steps that the rest of the guide takes one at a time.

1.1 What You Are Building

The TFR/911H is a single board computer with a Hitachi 6309 for its CPU and a Raspberry Pi RP2350B for everything else: the 6309's clock, its memory, and all of its I/O. The RP2350B talks USB to a program on your computer, the *tether*, which is the board's terminal.

A FujiNet is an ESP32 with WiFi and an SD card. Three wires join it to the TFR/911H, and over them the two speak DriveWire, the protocol a Color Computer uses to reach disks and the network. The FujiNet's SD card holds the disk that NitrOS-9 boots from; its WiFi reaches the servers that the network programs talk to. Figure 1–1 shows the whole system.

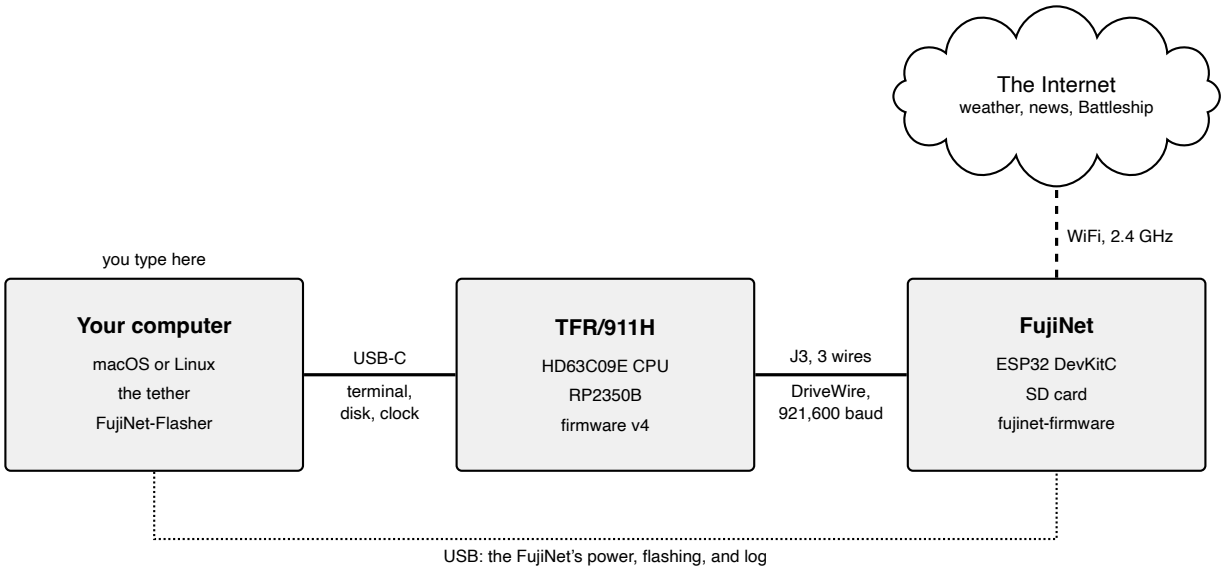


Figure 1–1 The TFR/911H, the FujiNet, and your computer

When you are done, the TFR/911H's firmware boots NitrOS-9 Level 2 from the file `level2-fn.dsk` on the FujiNet's SD card, which NitrOS-9 calls `/DD`, its system disk. You use it through the tether's window on your computer.

1.2 What Is in the Release

The release is a folder, `jamesBrown1/`, laid out as Figure 1–2 shows. Appendix A lists every file with its size and checksum.

There are two disks. `sdcard/tfr911/level2-fn.dsk` goes on the FujiNet's SD card and is the one NitrOS-9 boots from. `tether/level2.dsk` stays on your computer, where the tether serves it as a second hard disk, `/H0`; it can also boot NitrOS-9 by itself, without the FujiNet (Section 6.6). Both hold the same system and the same commands.

```
jamesBrown1/
  README.txt                       what this is, and the steps in brief
  TFR-Installation-Guide-jamesBrown1.pdf
                                   this guide
  TFR-Manual-and-Reference-1.3.pdf the TFR Manual and Reference
  SHA256SUMS                       a checksum for every file
  tfr911h/
    tfr911h_v4.uf2                 the TFR/911H's firmware                    (Chapter 4)
  fujinet/
    fujinet-COCO-HS-UART-v1.6.2-dev-dca3ee4a3.zip
                                   the FujiNet's firmware, for FujiNet-Flasher (Chapter 2)
  sdcard/                          what goes on the FujiNet's SD card         (Chapter 3)
    fnconfig.ini                   the FujiNet's settings: WiFi, host 1, device slot 1
    tfr911/
      level2-fn.dsk                NitrOS-9 Level 2, booting from the FujiNet
  tether/                          what stays on your computer                (Chapter 4)
    run-tfr.sh                     starts the tether and boots NitrOS-9
    tether-darwin-arm64            the tether for a Mac with Apple silicon
    tether-darwin-amd64            ... a Mac with an Intel processor
    tether-linux-amd64             ... a Linux PC
    tether-linux-arm64             ... 64-bit Linux on ARM: a Raspberry Pi 4 or 5
    tether-linux-arm               ... 32-bit Linux on ARMv7: a Raspberry Pi 2 or 3
    level2.dsk                     the tether's own disk, /H0
  licenses/
    LICENSE-tfr9-MIT.txt           the TFR/9 software's license
    LICENSE-GPL-3.0.txt            the FujiNet firmware's and programs' license
```

Figure 1–2 The release folder

1.3 What You Need

Table 1–1 What you need

Item	Notes
A TFR/911H	With its HD63C09E (or 6309E) in the socket.
An ESP32 board for the FujiNet	An Espressif ESP32-DevKitC with an ESP32-WROVER module, which has 8 MB of flash and PSRAM: the ESP32-DevKitC-VE, for one. The firmware needs both the 8 MB and the PSRAM; it will not run on a board with 4 MB of flash, or on an ESP32-WROOM module, which has no PSRAM. A FujiNet built for the Centipede also works.
An SD card, and a way to connect it	A microSD card of 32 GB or less. If your ESP32 board has no SD card slot, a microSD breakout board for 3.3 volts and six more jumper wires (Section 2.2).
Three jumper wires	Male at one end and female at the other, 15 cm (6 inches) long or less: the TFR/911H's header J3 is a socket, and the DevKitC's pins are pins. Yellow, green, and black make the wiring easy to check.
Two USB cables	One for the TFR/911H (USB-C), and one for the ESP32 board (usually micro-USB). Both must carry data, not only charge.
A computer	A Mac (macOS 11 or later, Apple silicon or Intel) or a Linux computer (a PC, or a Raspberry Pi 2 or later), with two free USB ports, an SD card reader, and a terminal of at least 80 columns by 24 lines that shows UTF-8. This release has no tether for Windows.
FujiNet-Flasher	The FujiNet project's flashing program, version 1.5.0 or later, from `github.com/FujiNetWIFI/fujinet-flasher/releases`.
A WiFi network	2.4 GHz, for the network programs. NitrOS-9 boots from the SD card without it.

1.4 The Installation at a Glance

Figure 1–3 shows the six steps. The FujiNet comes first, so that it is ready when the TFR/911H first asks it for something.

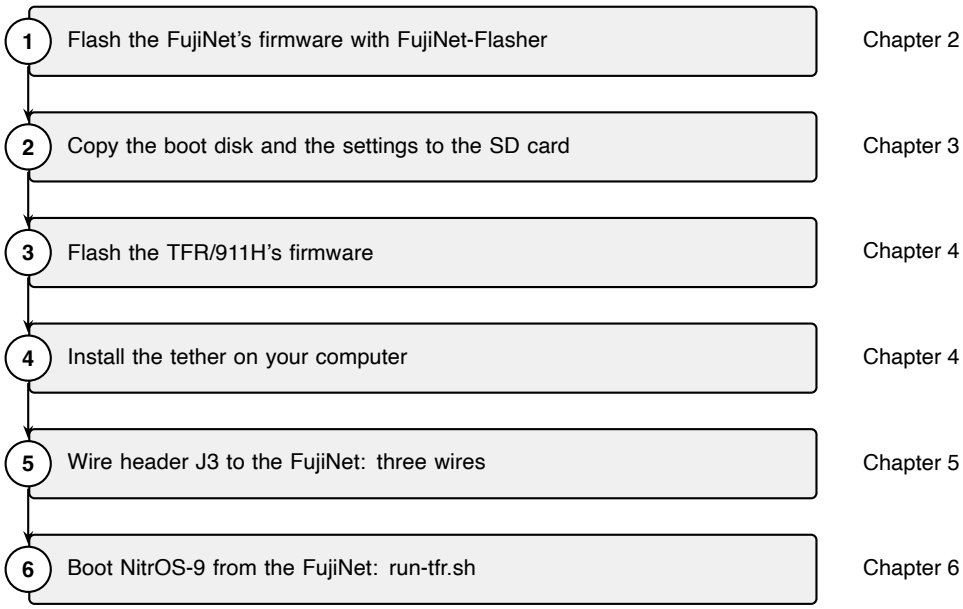


Figure 1–3 The installation, step by step

Flashing the FujiNet 2

This chapter describes the ESP32 board the FujiNet runs on, how to give it an SD card if it has none, and how to flash its firmware with FujiNet-Flasher.

2.1 The ESP32 Board

The FujiNet firmware in the release, `fujinet/fujinet-COCO-HS-UART-v1.6.2-dev-dca3ee4a3.zip`, is *fujinet-firmware* built for the board `fujinet-coco-devkitc-centipede`: the Color Computer's FujiNet, speaking DriveWire over a plain 3.3-volt serial link at a fixed 921,600 baud on the ESP32's UART1, which is what the TFR/911H expects. It was made for an ESP32-DevKitC, and Figure 2–1 shows the pins it uses on one.

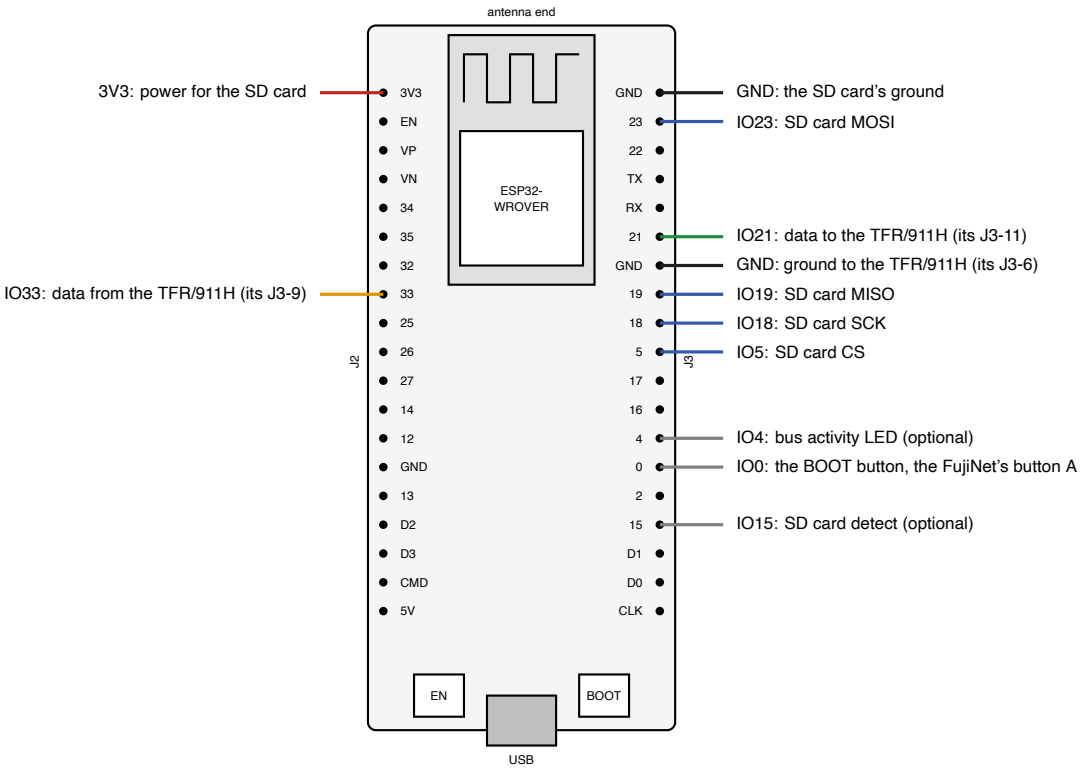


Figure 2–1 An ESP32-DevKitC, seen from above, and the pins the FujiNet uses

The pins that matter most are the three that go to the TFR/911H: IO33 (the FujiNet's receive data), IO21 (its transmit data), and a ground. Chapter 5 wires them. The DevKitC's BOOT button is the FujiNet's button A; its EN button restarts the ESP32.

A DevKitC has no Safe Reset button, the FujiNet's button C, and the firmware in this release leaves it out. Older firmware gave the button IO14, which a DevKitC leaves unconnected; the floating pin picked up the UART beside it and restarted the FujiNet every few seconds under disk traffic. That was fixed in fujinet-firmware pull request 2034, which this firmware includes.

2.2 Adding an SD Card

If your ESP32 board has no SD card slot, wire a microSD breakout board to it as Table 2–1 shows. Use a breakout made for 3.3 volts, or one with its own regulator and level shifters that takes 3.3 volts in. The card-detect wire is optional: without it, the FujiNet looks for the card only when it starts.

Table 2–1 Wiring a microSD breakout to the DevKitC

Breakout	DevKitC pin	Header	Signal
CS	IO5	J3-10	Chip select
SCK (CLK)	IO18	J3-9	Clock
MOSI (DI)	IO23	J3-2	Data to the card
MISO (DO)	IO19	J3-8	Data from the card
VCC (3V3)	3V3	J2-1	Power
GND	GND	J3-1	Ground
CD	IO15	J3-16	Card detect, if the breakout has it

2.3 Installing FujiNet-Flasher

FujiNet-Flasher is the FujiNet project's program for flashing a FujiNet. It needs no installing: download the file for your computer from `github.com/FujiNetWIFI/fujinet-flasher/releases`, unzip it, and start it.

macOS — `FujiNet-Flasher_v1.5.0_macos-10.15+_unsigned.zip`. The program is not signed, so the first time, hold down Control as you click it, choose Open, and then Open again. If macOS still will not open it, open System Settings, choose Privacy & Security, and click Open Anyway.

Linux — `FujiNet-Flasher_v1.5.0_linux-x64.zip`. To use the serial port, your user must be in the group that owns it: **`sudo usermod -aG dialout $USER`**, then log out and in again (on Arch Linux the group is `uucp`).

The DevKitC's USB port is a USB-to-serial bridge, usually a Silicon Labs CP2102N or a WCH CH340. If no serial port appears when you plug the board in, install that chip's driver, or try another cable: many USB cables carry only power.

2.4 Flashing the Firmware

FujiNet-Flasher can flash an official FujiNet release, which it downloads, or a firmware file you give it. Give it the file from the release.

Caution Do not choose an official release. The Tandy CoCo releases are for a Color Computer's FujiNet, which talks to the CoCo at another speed on other pins; flashed with one, the FujiNet cannot talk to the TFR/911H.

1. Plug the DevKitC's USB port into your computer. Leave it unplugged from the TFR/911H for now.
2. Start FujiNet-Flasher. Figure 2–2 shows its window.
3. In Serial port, choose the DevKitC's port. On macOS it is `/dev/cu.usbserial-`*n* or `/dev/cu.SLAB_USBtoUART`; on Linux, `/dev/ttyUSB0`. If it is not in the list, click the button beside it, which reads the list again.
4. Leave Baud Rate at 460800.
5. In Custom Firmware File, click the button and choose `fujinet/fujinet-COCO-HS-UART-v1.6.2-dev-dca3ee4a3.zip` from the release. Firmware then says `Custom Firmware File Selected`.
6. Click Flash FujiNet Firmware.

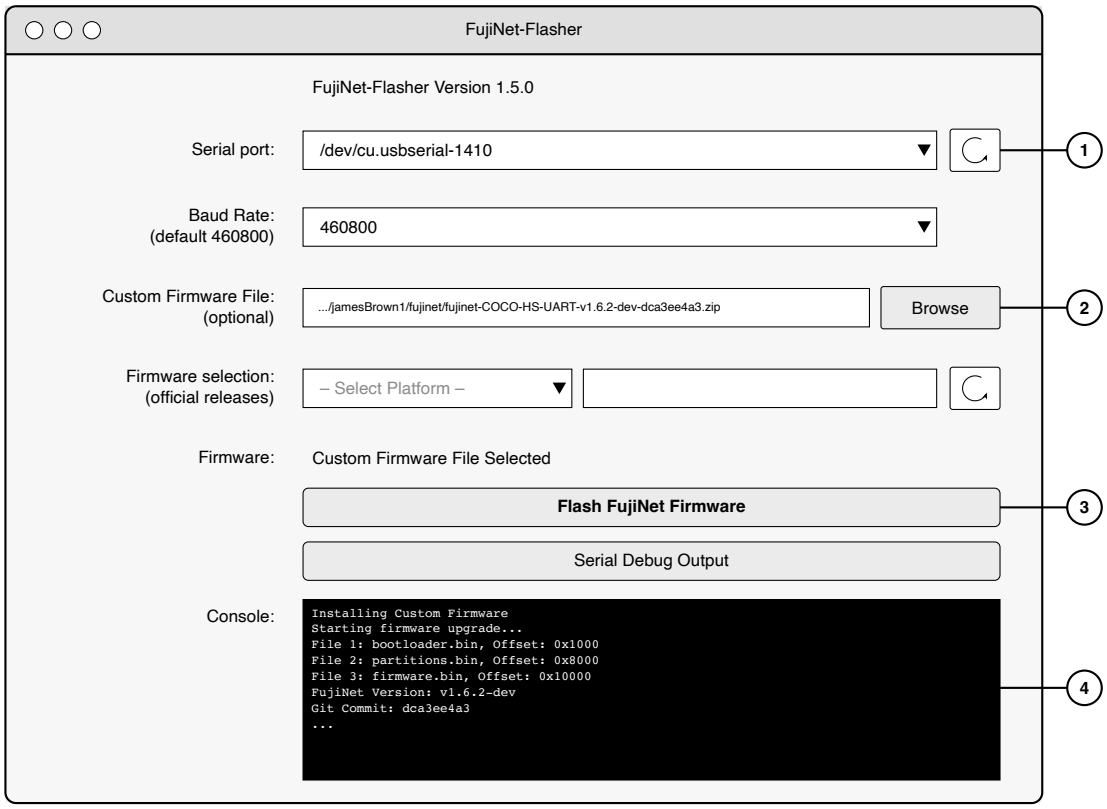


Figure 2–2 FujiNet-Flasher, ready to flash: (1) the DevKitC's serial port; (2) the firmware from the release; (3) the button that flashes it; (4) its progress, and then the FujiNet's log

FujiNet-Flasher erases the whole of the ESP32's flash, writes the five parts of the firmware—`bootloader.bin`, `partitions.bin`, `firmware.bin`, `update.bin`, and `littlefs.bin`, the web pages—restarts the ESP32, and shows its log in the console. It takes a few minutes. At the end the console says:

```
Hard Resetting...
Done! Flashing is complete!
```

Note Erasing the flash also erases the settings the FujiNet keeps there. The copy of its settings on its SD card, `fnconfig.ini`, is not touched, and the FujiNet reads it again when it starts.

If FujiNet-Flasher cannot find the ESP32 (`Error Detecting ESP`), hold down the DevKitC's BOOT button, press and let go of EN, let go of BOOT, and click Flash FujiNet Firmware again. A lower baud rate, such as 115200, sometimes helps with a long cable.

2.5 Checking the Firmware

The console shows the FujiNet's log as it starts; Serial Debug Output shows it again whenever you click it (and restarts the ESP32). It should include these lines:

```
FujiNet v1.6.2-dev 2026-06-18 09:35:40 (COCO) Started @ 9
LittleFS mounted.
[lib/bus/drivewire/drivewire.cpp:827] readBaudSwitch(): COCO_HS_UART build, defaulting to 921600 baud
```

The first says which firmware it is; the last, that it is the build for the TFR/911H. Without an SD card it also says that it found none; you make the card in the next chapter.

Close FujiNet-Flasher before you go on: only one program at a time can have the serial port open, and opening it restarts the ESP32.

Making the SD Card 3

The FujiNet's SD card holds the disk that NitrOS-9 boots from and the FujiNet's settings. This chapter describes how to make it on your computer, with a card reader. Section 3.5 tells how to do the same over WiFi instead.

3.1 Formatting the Card

Use a microSD card of 32 GB or less, formatted FAT32. A new card usually is. To format one:

macOS	In Disk Utility, choose View, then Show All Devices; choose the card (the device, not a volume on it), click Erase, and choose the format MS-DOS (FAT) and the scheme Master Boot Record. Name it `FUJINET`.
Linux	Find the card's device with **`lsblk`**, then, if it is `/dev/sdb` with one partition: **`sudo mkfs.vfat -F 32 -n FUJINET /dev/sdb1`**. Check the device name twice; this erases it.

3.2 Copying the Files

Copy everything in the release's `sdcard/` folder to the top of the card, keeping the folder `tfr911`. Figure 3–1 shows the card when you are done.

```
$ cd jamesBrown1
$ cp -R sdcard/ /Volumes/FUJINET/                (macOS)
$ cp -R sdcard/. /media/$USER/FUJINET/           (Linux)
```

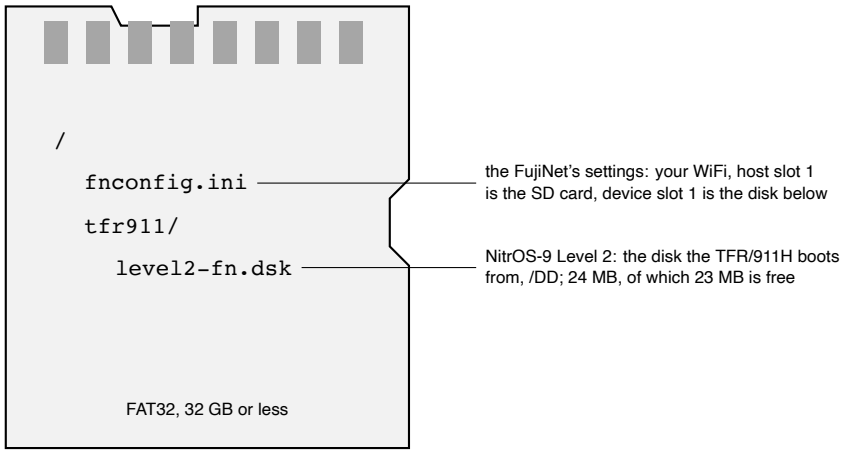


Figure 3–1 The FujiNet's SD card

The disk image is about 500 KB as a file, but it is a 24 MB disk to NitrOS-9: the file grows as NitrOS-9 fills it.

3.3 The FujiNet's Settings

`fnconfig.ini` holds the FujiNet's settings. The one in the release sets three things: your WiFi network, host slot 1 (the SD card), and device slot 1 (the boot disk). Open it in a text editor (TextEdit on a Mac, in plain text) and put your network's name and password in place of `YOUR-NETWORK` and `YOUR-PASSWORD`:

```
[WiFi]
enabled=1
SSID=your network's name
passphrase=its password
```

The rest of the file is set already:

`[General]`	The FujiNet's name on your network, `fujinet-tfr911`; `encrypt_passphrase=0`, so that the password above is read as you typed it.
`[Host1]`	Host slot 1 is the SD card. `[Host2]` is the FujiNet project's TNFS server, `fujinet.online`.
`[Mount1]`	Device slot 1 holds `/tfr911/level2-fn.dsk`, from host slot 1, read and write (`mode=w`). Device slot 1 is DriveWire drive 0, `/X0` to NitrOS-9, and it is the slot the TFR/911H boots from.

Appendix C lists every setting the file uses. The FujiNet reads the file from the SD card each time it starts, copies it to its flash, and writes both copies again when its settings change, without the notes at the top.

Note The password is stored on the card as plain text. Anyone with the card can read it.

To boot without WiFi, change `enabled=1` in the `[WiFi]` section to `enabled=0`. NitrOS-9 boots and runs as well, but the network programs have nothing to talk to. You can set the WiFi later from NitrOS-9 (Section 6.5).

3.4 Putting the Card in the FujiNet

Eject the card from your computer, put it in the FujiNet's slot, and press the DevKitC's EN button (or unplug its USB and plug it in again). If you watch its log (Serial Debug Output in FujiNet-Flasher), it says:

```
SD mounted.
Load fnconfig.ini from SD
...
WiFi connect attempt to SSID "your network's name"
...
Obtained IP address: 192.168.1.232
Starting web server on port 80
```

Write down the IP address. With it you can open the FujiNet's web page (`http://`*address*`/`) and reach its SD card over WiFi.

3.5 Doing It over WiFi

Once the FujiNet is on your network, its web server offers its SD card by WebDAV at `/dav`. To copy the boot disk to it from your computer, with *address* the FujiNet's IP address:

```
$ curl -X MKCOL http://address/dav/tfr911
$ curl -T sdcard/tfr911/level2-fn.dsk http://address/dav/tfr911/level2-fn.dsk
```

Then put the disk in device slot 1, read and write, on the FujiNet's web page, or from NitrOS-9 booted from the tether's disk (Section 6.6):

```
OS9: fnsetdevfile 0 1 2 /tfr911/level2-fn.dsk
```

(**fnsetdevfile** counts device slots from 0: slot 0 is the FujiNet's slot 1. Host slot 1 is the SD card, and mode 2 is read/write.)

The same way, **`curl -o level2-fn-backup.dsk http://`*`address`*`/dav/tfr911/level2-fn.dsk`** copies the disk back to your computer, to keep a backup.

Preparing the TFR/911H 4

This chapter describes how to flash the TFR/911H's firmware and install the tether, and how to check that they work before the FujiNet is connected.

4.1 The Board

Figure 4–1 shows the parts of the TFR/911H this guide uses: the USB-C connector, through which the board takes its power and talks to the tether; the RESET and FLASH buttons; and header J3, the 20-pin socket at the lower left, where the FujiNet connects.

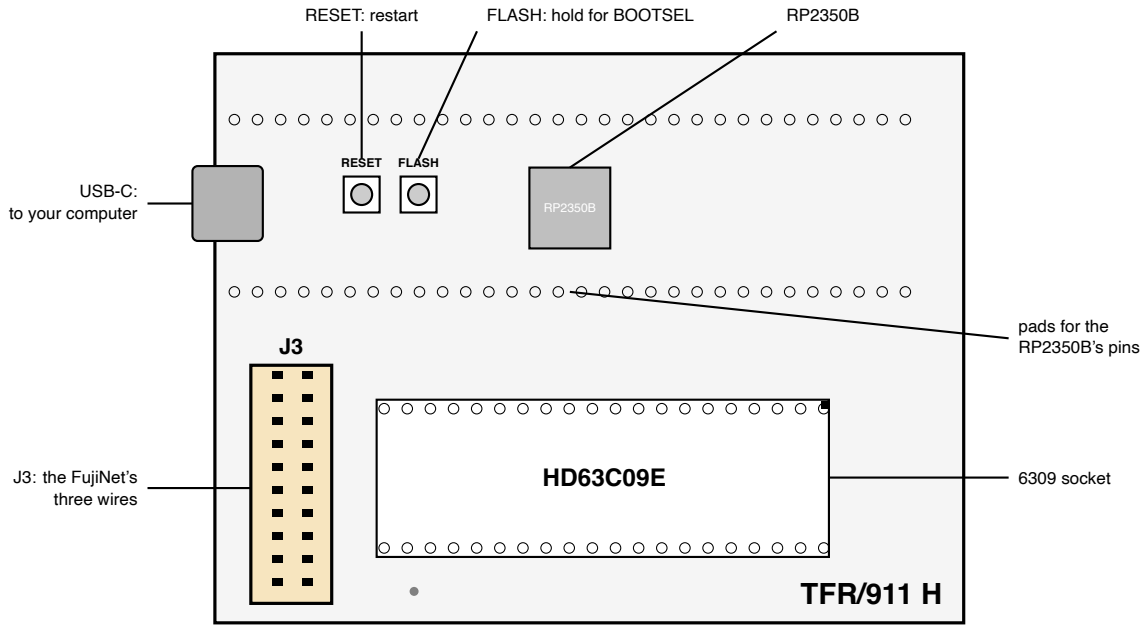


Figure 4–1 The TFR/911H, seen from above

4.2 Flashing the Firmware

The TFR/911H's firmware is `tfr911h/tfr911h_v4.uf2`. It holds the firmware's console (the `TCL>` prompt), its two NitrOS-9 kernels, one booting from the tether's disk (**l2**) and one from the FujiNet (**l2fn**), and the becker port through which the 6309 talks to the FujiNet.

The RP2350B has a boot loader in ROM, BOOTSEL, which makes the board look like a USB drive. Copying a `.uf2` file to that drive flashes the firmware. Figure 4–2 shows how to start BOOTSEL with the board's two buttons.

1. Plug the TFR/911H into your computer with the USB-C cable.
2. Hold down FLASH. Press and let go of RESET. Then let go of FLASH. (Or hold down FLASH while you plug the cable in.)

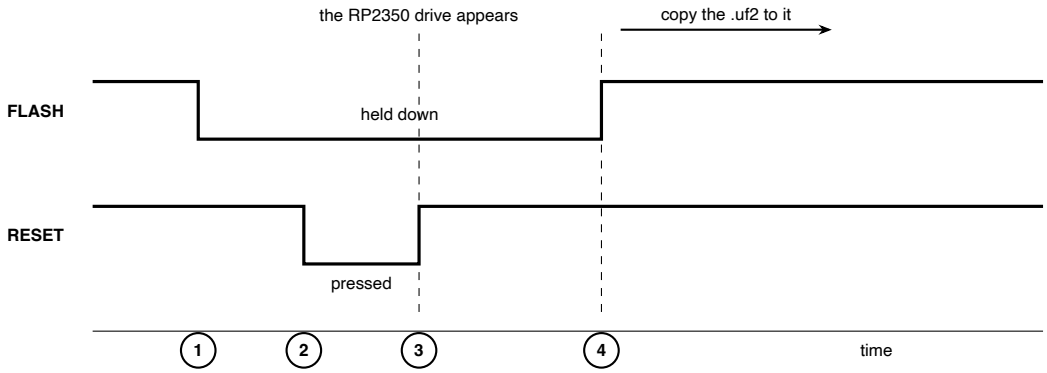


Figure 4–2 Starting BOOTSEL: (1) hold down FLASH; (2) press RESET; (3) let go of RESET; (4) let go of FLASH

3. A drive named `RP2350` appears on your computer.
4. Copy `tfr911h_v4.uf2` to it, by dragging it in the Finder or the file manager, or:

```
$ cp tfr911h/tfr911h_v4.uf2 /Volumes/RP2350/          (macOS)
$ cp tfr911h/tfr911h_v4.uf2 /media/$USER/RP2350/      (Linux)
```

5. The drive goes away by itself and the board restarts with the new firmware. macOS may complain that the disk was not ejected properly; that is expected, and does no harm.

With Raspberry Pi's **picotool**, **`picotool load -x tfr911h/tfr911h_v4.uf2`** does steps 4 and 5, once the board is in BOOTSEL.

Caution Quit the tether before you flash the firmware: only one program at a time can have the board's serial port open.

4.3 Installing the Tether

The tether is a program on your computer that talks to the TFR/911H over USB. It is the board's terminal: what you type goes to the 6309, and what the 6309 prints appears in the tether's window. It also serves a disk image on your computer to NitrOS-9 as `/H0`, and sets the board's clock when it connects.

Copy the release's `tether/` folder anywhere you like; it holds the tether for each kind of computer (Table 4–1), the disk it serves, and `run-tfr.sh`, which starts it.

Table 4–1 The tethers

File	For
`tether-darwin-arm64`	A Mac with Apple silicon (M1 and later)
`tether-darwin-amd64`	A Mac with an Intel processor
`tether-linux-amd64`	A Linux PC (x86-64)
`tether-linux-arm64`	64-bit Linux on ARM: a Raspberry Pi 3, 4, or 5 with a 64-bit system
`tether-linux-arm`	32-bit Linux on ARMv7: a Raspberry Pi 2, 3, or 4 with a 32-bit system

`run-tfr.sh` chooses the right one. Before you run it the first time:

macOS	Files downloaded from the Internet are quarantined, and macOS will not run a quarantined program that is not signed. Take the release out of quarantine: **`xattr -dr com.apple.quarantine jamesBrown1`**
Linux	Your user must be in the group that owns the board's serial port, `/dev/ttyACM0`: **`sudo usermod -aG dialout $USER`**, then log out and in again (on Arch Linux, the group `uucp`).

The tether runs in a terminal window: Terminal or iTerm2 on a Mac, any terminal on Linux. Make the window at least 80 columns by 24 lines; the NitrOS-9 programs in the release draw full-screen, in color, with the box-drawing characters of UTF-8, which the usual terminal fonts (Menlo, SF Mono, DejaVu Sans Mono) all have.

4.4 First Contact

To check the firmware and the tether, run `run-tfr.sh` with `none`, which connects without booting anything:

```
$ cd jamesBrown1/tether
$ sh run-tfr.sh none
TFR/911H on /dev/cu.usbmodem13401; disk .../level2.dsk; boot nothing.
^C goes to OS-9.  Quit with ^] .

TCL> fujinet
no answer from the FujiNet (0 of 6 bytes; GPIO 12 TX, 13 RX, 921600 baud)
TCL>
```

If `TCL>` does not appear at once, press Return. The **fujinet** command asks the FujiNet the time; with no FujiNet connected yet, it has no answer. Quit the tether with Ctrl/].

`run-tfr.sh` finds the board's serial device by itself when there is only one: `/dev/cu.usbmodem*` on a Mac, `/dev/ttyACM*` on Linux. If there are several, it lists them; set `WIRE` to the right one:

```
$ WIRE=/dev/cu.usbmodem13401 sh run-tfr.sh none
```

Connecting the FujiNet 5

This chapter describes how to wire the FujiNet to the TFR/911H with three wires, how to power the two boards, and how to check the link between them.

5.1 Header J3

The FujiNet connects to header J3, the 2-by-10 socket at the lower left of the TFR/911H, beside the 6309 (Figure 4–1). The name of each pin is printed beside it; Figure 5–1 shows them. Pin 1, marked G8, is at the bottom left; the odd pins run up the left column and the even pins up the right.

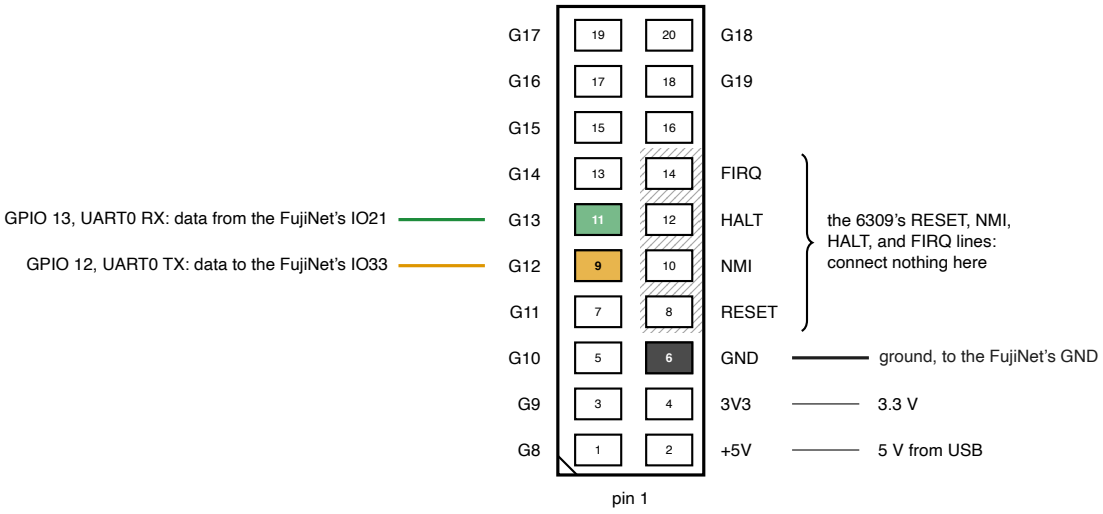


Figure 5–1 Header J3, seen from above, with the 6309 to the right

Three of J3's pins carry the link to the FujiNet: G12 (pin 9), G13 (pin 11), and GND (pin 6).

Caution Pins 8, 10, 12, and 14 are wired to the 6309's RESET, NMI, HALT, and FIRQ lines. Connect nothing to them. Do not connect the 3V3 or +5V pins to the FujiNet either: each board takes its power from its own USB cable.

5.2 Wiring

Figure 5–2 and Table 5–1 show the three wires. Each data wire goes from one board's transmit pin to the other's receive pin; the table has that crossing in it already, so wire it just as it says.

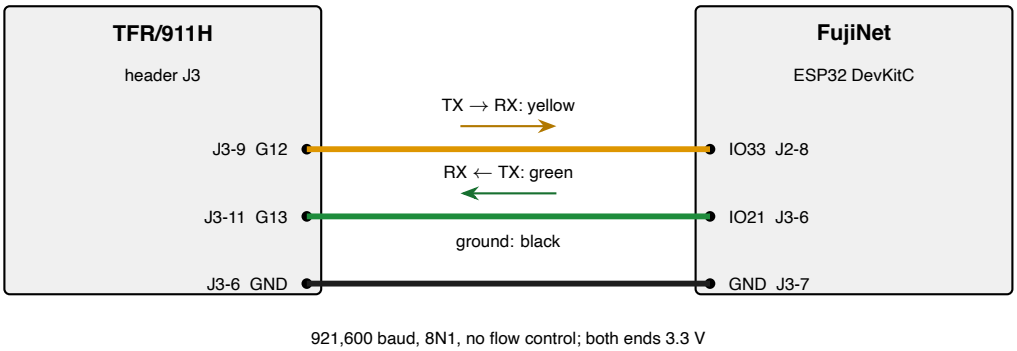


Figure 5–2 The three wires between the TFR/911H and the FujiNet

Table 5–1 The wires

TFR/911H	Signal	FujiNet	Signal	Wire
J3-9, G12	GPIO 12, UART0 TX	IO33, J2-8	GPIO 33, UART1 RX	yellow
J3-11, G13	GPIO 13, UART0 RX	IO21, J3-6	GPIO 21, UART1 TX	green
J3-6, GND	ground	GND, J3-7	ground	black

1. Unplug both boards from USB.
2. Push the female end of the yellow wire onto the DevKitC's IO33, the eighth pin down on the left, counting from the antenna end (Figure 2–1). Push its male end into J3 pin 9, G12.
3. Push the female end of the green wire onto the DevKitC's IO21, the sixth pin down on the right. Push its male end into J3 pin 11, G13.
4. Push the female end of the black wire onto the DevKitC's GND, the seventh pin down on the right. Push its male end into J3 pin 6, GND.
5. Check each wire against Table 5–1 before you plug anything in.

Any of the DevKitC's GND pins will do for the ground wire. Keep the wires short, 15 cm (6 inches) or less: the link runs at 921,600 baud.

5.3 Power

Each board takes its power from its own USB cable, and the ground wire joins the two. Plug the FujiNet in first, so that it has started—a few seconds—before the TFR/911H first asks it for something; then the TFR/911H.

The FujiNet's USB cable can go to your computer, where it also carries the FujiNet's log, or to any USB charger.

Caution Keep the FujiNet powered while NitrOS-9 runs from it. NitrOS-9 waits for each answer from the FujiNet with nothing else running, and a FujiNet that stops answering stops NitrOS-9 with it. If that happens, press RESET on the TFR/911H and boot again.

5.4 Checking the Link

Run `run-tfr.sh` with `none` again, and at the `TCL>` prompt try the **fujinet** commands:

```
$ sh run-tfr.sh none
TCL> fujinet
FujiNet says it is 2026-10-20 19:41:07
TCL> fujinet mount 1
```

```
slot 1: mounted level2-fn.dsk
TCL> fujinet stats
rx n tx n rx_drops 0 tx_drops 0 overruns 0 empty_reads 0
```

fujinet	Asks the FujiNet the time (DriveWire's `OP_TIME`) and prints its answer. An answer proves both data wires, the ground, and the baud rate. The time is the FujiNet's own, which it sets from the network; without WiFi it starts in 1970, which does no harm.
fujinet mount 1	Mounts the disk in device slot 1, as booting will. It proves the SD card, `fnconfig.ini`, and the disk.
fujinet stats	Counts the bytes each way, and any lost. The four error counts should stay at 0.

If **fujinet** has no answer, see Chapter 7.

Booting from the FujiNet 6

This chapter describes how to boot NitrOS-9 from the FujiNet, what happens when you do, where everything is afterwards, and what to try first.

6.1 Booting

With the FujiNet powered and connected, run `run-tfr.sh` with no argument. It starts the tether and types **l2fn** at the `TCL>` prompt for you:

```
$ cd jamesBrown1/tether
$ sh run-tfr.sh
TFR/911H on /dev/cu.usbmodem13401; disk .../level2.dsk; boot l2fn.
^C goes to OS-9.  Quit with ^] .
[Injecting: l2fn]
FujiNet slot 1: level2-fn.dsk
[L2: starting 6309 with NitrOS-9 Level 2]
[bye: asserting RESET+HALT...]
   ...
[bye: Phase 2 started]
L2FN
NitrOS-9/6809 Level 2 V03.3.0
Tandy Color Computer 3
(C) 2014 The NitrOS-9 Project
Tue Oct 20 19:42:16 2026
http://www.nitros9.org

NitrOS-9 Level 2 on the TFR/911H.  Unix-style commands: man unix
Shell

OS9:
```

The lines in brackets are the firmware's, as it hands the board to the 6309; `L2FN` is the first thing the 6309 prints. NitrOS-9 is running, from the FujiNet's SD card. (`Tandy Color Computer 3` is the machine NitrOS-9 was built for; the TFR/911H's firmware presents the 6309 with what NitrOS-9 needs of one.)

At `TCL>` you can also type **l2fn** yourself, after **`sh run-tfr.sh none`**.

6.2 What Happens

Figure 6–1 shows what happens between **l2fn** and the `OS9:` prompt.

1. The tether connects to the board and sets its clock. Given `-boot l2fn`, it types **l2fn** at the `TCL>` prompt.
2. When the FujiNet starts, it puts its own CONFIG disk in DriveWire drive 0. So **l2fn** first asks the FujiNet which disk device slot 1 names, mounts that slot's host (the SD card), then the disk, read and write, and prints `FujiNet slot 1: level2-fn.dsk`.
3. The firmware copies NitrOS-9's kernel into the 6309's memory from its own flash, and starts the 6309.

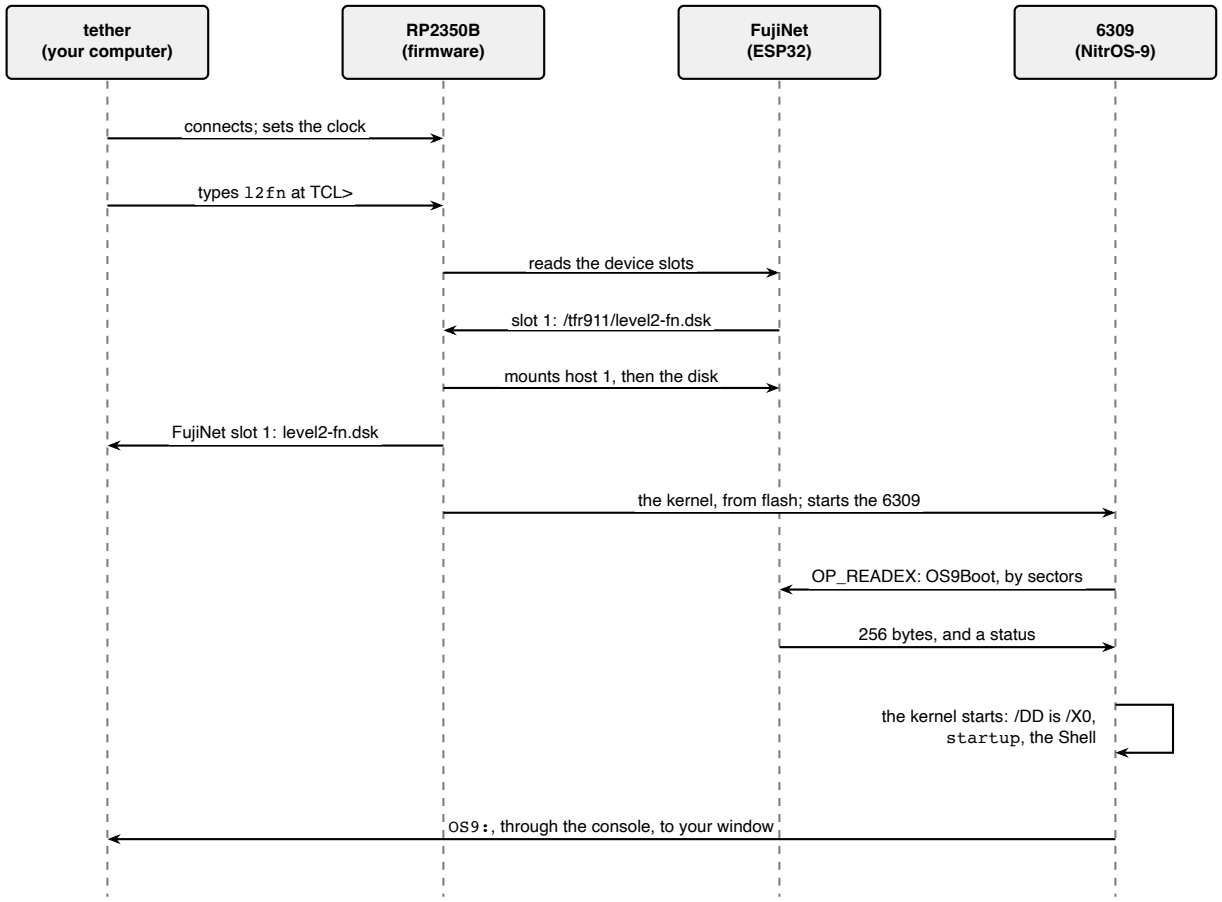


Figure 6–1 Booting from the FujiNet

4. The kernel's `Boot` module, `boot_dw`, reads the file `OS9Boot` from DriveWire drive 0, a sector at a time, through the becker port at `$FF45`–`$FF46` that the firmware makes of the link.
5. `OS9Boot` names `/X0`, the FujiNet's slot 1, as `/DD`, the system disk. NitrOS-9 runs `startup` from it, then the Shell, whose `OS9:` prompt comes to your window through the tether.

6.3 Where Everything Is

Figure 6–2 shows what NitrOS-9's devices are, booted from the FujiNet.

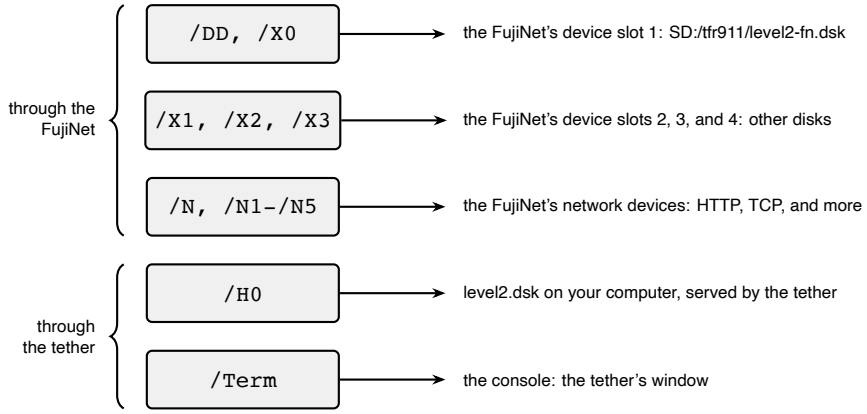


Figure 6–2 NitrOS-9's devices, booted from the FujiNet

Everything you save in `/DD` goes to `level2-fn.dsk` on the FujiNet's SD card. `/H0` is a second disk with the same system on it, on your computer: **`dir /h0`** shows it, and **`copy /h0/`***file* **`/dd/`***file* copies from it.

6.4 Trying It Out

fnstatus	The FujiNet's firmware version, its WiFi network, and its IP address.
fnweather	The weather where you are, now and for the week, with pictures.
fnnews	The news.
fnbattleship	Fuji Battleship: sea battle for up to four players, on the Internet, in color, with sound if your board has a speaker.
fnbounce	Bounce World: shapes bouncing from one computer's screen to the next.
fnstocks	Ten stock quotes and a ticker.
info	The system and the hardware.
man unix	The Unix-style commands: **ls**, **cat**, **cp**, **vi**, and more.

The *TFR Manual and Reference*, in the release, describes these and every other command.

6.5 Setting the WiFi from NitrOS-9

If you did not put your WiFi network in `fnconfig.ini`, or it has changed, set it from NitrOS-9 with **fnconfig**:

1. Type **`fnconfig`**, and press `4` for its WiFi page.
2. Press `S` to scan for networks and choose yours, or `E` to type its name.
3. Type its password. The FujiNet joins the network and saves the setting, on its SD card and in its flash.
4. Press `1` for the status page, which shows the FujiNet's IP address, and `Q` to quit.

Booted from the FujiNet, **fnconfig** asks before it changes the WiFi: changing it drops the network connection for a moment. The disk does not depend on the WiFi, since it is on the SD card.

6.6 Booting from the Tether's Disk

The TFR/911H can also boot NitrOS-9 from `tether/level2.dsk` on your computer, without the FujiNet:

```
$ sh run-tfr.sh l2
```

Then `/DD` is `/H0`, the tether's disk. If the FujiNet is connected, its disks are still `/X0`–`/X3`, and the network programs still work. This is the way to set up a FujiNet over WiFi before its SD card has a boot disk (Section 3.5), and a way to start the system when the FujiNet is not there.

6.7 Coming and Going

- Ctrl/C goes to NitrOS-9: it stops the running command. Ctrl/] quits the tether.
- Quitting the tether does not stop NitrOS-9; the board keeps running. **`sh run-tfr.sh`** again connects to it as it is (it boots nothing when the 6309 is already running).
- To start over, press RESET on the TFR/911H, or unplug it, and run `run-tfr.sh` again.
- Before you take the SD card out of the FujiNet, or replace `level2-fn.dsk` on it, stop NitrOS-9: quit the tether, then press RESET. The FujiNet keeps the disk file open while it is mounted.
- After you replace the disk on the card, **l2fn** (which mounts the disk again) or **`fujinet mount 1`** makes the FujiNet open the new file.

Troubleshooting 7

Table 7–1 lists what can go wrong, step by step, and what to do about it. The FujiNet's log, which Serial Debug Output in FujiNet-Flasher shows, and the tether's log, `tether/tether.log`, often say what happened.

Table 7–1 Troubleshooting

Symptom	Cause	What to do
Flashing the FujiNet		
FujiNet-Flasher lists no serial port for the DevKitC.	No driver for its USB bridge, a cable that only charges, or no permission.	Install the CP210x or CH340 driver; try another cable; on Linux, join the `dialout` group. Click the button beside Serial port to read the list again.
`Error Detecting ESP.`	The ESP32 did not enter its download mode, or another program has the port.	Close any serial monitor. Hold BOOT, press EN, let go of BOOT, and flash again; try 115200 baud.
`Firmware larger than chip flash.`	The board has 4 MB of flash.	Use a board with 8 MB.
After flashing, the FujiNet restarts over and over, and its log speaks of PSRAM or SPI RAM.	The module has no PSRAM (an ESP32-WROOM).	Use a board with an ESP32-WROVER module.
The log has no `COCO_HS_UART` line.	An official release was flashed, not the release's zip.	Flash `fujinet-COCO-HS-UART-v1.6.2-dev-dca3ee4a3.zip` (Chapter 2).
The SD card		
The log says the FujiNet found no SD card.	The card is not FAT32, not seated, or not wired.	Format it FAT32 (Chapter 3); check the breakout's wiring against Table 2–1.
The FujiNet does not join the WiFi.	Wrong name or password, or a 5 GHz network.	Check `SSID` and `passphrase` in `fnconfig.ini`, which are case sensitive; use a 2.4 GHz network; or set it with **fnconfig** (Section 6.5).
The TFR/911H and the tether		
No `RP2350` drive appears.	The board did not enter BOOTSEL, or the cable only charges.	Hold FLASH while you press and let go of RESET; try another cable.
`Plug in the TFR/911H, or set WIRE to its serial device.`	No board found, or more than one candidate.	Plug it in; or set `WIRE` to the device it lists that appears when you plug the board in.
macOS: `cannot be opened because the developer cannot be verified.`	The release is quarantined.	**`xattr -dr com.apple.quarantine jamesBrown1`**
Linux: `permission denied` on `/dev/ttyACM0`.	Your user is not in the serial port's group.	**`sudo usermod -aG dialout $USER`**; log out and in.
The tether exits at once, or says the port is busy.	Another program, or another tether, has the port.	Quit it. Only one program can have the port open.
Boxes and lines show as odd characters; colors are missing.	The terminal is not set for UTF-8, or its font lacks the characters.	Set the terminal to UTF-8; use Menlo, SF Mono, or DejaVu Sans Mono.
The link and booting		

Symptom	Cause	What to do
`no answer from the FujiNet (0 of 6 bytes ...)`.	A wire is missing or swapped, there is no ground, the FujiNet is off or still starting, or it has the wrong firmware.	Check the wires against Table 5–1: J3-9 to IO33, J3-11 to IO21, J3-6 to GND. Wait for the FujiNet to start. Check its log for the `COCO_HS_UART` line.
`FujiNet slot 1: nothing is in that device slot.`	Device slot 1 is empty: `fnconfig.ini` is not on the card, or has no `[Mount1]`.	Copy `fnconfig.ini` to the top of the card (Chapter 3), or set the slot with **fnsetdevfile** or the FujiNet's web page.
`FujiNet slot 1: the FujiNet could not mount the image.`	The disk is not where the slot says.	Check that the card has `/tfr911/level2-fn.dsk`, spelled just so.
The boot stops after `L2FN`, or NitrOS-9 stops while it runs.	The FujiNet restarted or lost power, or the link is unreliable.	Press RESET and boot again. Check `fujinet stats` at `TCL>` for lost bytes; shorten the wires; make sure the FujiNet has this release's firmware.
The network programs fail, and **fnstatus** shows no IP address.	The FujiNet is not on the WiFi.	Set the WiFi (Section 6.5).

About This Release A

A.1 The Files

Table A–1 lists the files of release jamesBrown1 with their sizes and SHA-256 checksums. `SHA256SUMS`, in the release, lists the same and the rest (this guide, the manual, `README.txt`, and the licenses). To check them all:

```
$ cd jamesBrown1
$ shasum -a 256 -c SHA256SUMS          (macOS; on Linux, sha256sum -c SHA256SUMS)
```

Table A–1 The files of release jamesBrown1

File, and its SHA-256	Bytes
`tfr911h/tfr911h_v4.uf2`	1,356,800
`12f6f19e594f3b25570ad23374d3cbe888bac800f3e094e46fdab670e5e03388`	
`fujinet/fujinet-COCO-HS-UART-v1.6.2-dev-dca3ee4a3.zip`	6,286,854
`dcb995801d05ef61c553bf4f9647c8697661db51c4912eb0b934b632f88afd50`	
`sdcard/fnconfig.ini`	933
`0d40009da101ac7f6b175eff7aea3e09323d1e94021aa9146153f2489dc43228`	
`sdcard/tfr911/level2-fn.dsk`	505,088
`374e5a44082d374a5b63ef4720759ec78b2012a94cfde3e1192ba760d0d7a553`	
`tether/level2.dsk`	484,608
`c99af8d73e10ae389991adbe0e9e65c058fa0c23afc69077d20be07844565077`	
`tether/run-tfr.sh`	2,316
`b8deafd1ae29d48b0005a046010d777a5251d196f3f7b8508f2cfacaf08a11f3`	
`tether/tether-darwin-arm64`	6,598,050
`51f6b70acc4aa4f0f68cf806b0faf22e1de02ead3503c62fb81c76306708cada`	
`tether/tether-darwin-amd64`	7,038,528
`d614e9a7859da87f2fb414b09d48a56c3335fdfc96de535c4a6412c8c71ea651`	
`tether/tether-linux-amd64`	6,926,520
`02f55d45d3b5b160638360c5830d547cc8de5f50de52d52b79f47d9aa9a47757`	
`tether/tether-linux-arm64`	6,553,784
`8947e2b737bdcb523024448ed21f3f05af7c21cb1f5ee10c0a20bde369acd98f`	
`tether/tether-linux-arm`	6,815,928
`04d5eb30632d489f1c74e780b02b2f92ffdfe89cad4681e9d979d92760708999`	

A.2 What Is in Each

TFR/911H firmware	v4, built 20 October 2026 from `dillera/tfr9`, branch `nitros9-l2-tfr911`, commit `77ad30c`, with pico-sdk 2.3.0 and the Arm GNU Toolchain 14.2. It holds TurbOS, the firmware's console, and the two NitrOS-9 Level 2 kernels.
NitrOS-9 disks	NitrOS-9/6809 Level 2 V3.3.0, built from `dillera/nitros9` commit `5afdac58` by the same `tfr9` commit, with the TFR/911H's commands and drivers, the FujiNet commands, and the C programs: **fnweather**, **fnnews**, **fnstocks**, **fnbounce**, **fnbattleship**, **tetris**, **zm**, **vi**, and **info**. The two disks differ only in their `OS9Boot` and boot track.
FujiNet firmware	fujinet-firmware v1.6.2-dev, built 20 October 2026 for the board `fujinet-coco-devkitc-centipede` from `FujiNetWIFI/fujinet-firmware` commit `dca3ee4a3`, "[coco] Disable BUTTON_C on DevKitC pin map (#2034)", by its own `build.sh -z`, which makes the zip that FujiNet-Flasher reads. It includes pull request 2027, which reads the version byte NitrOS-9 sends after `OP_DWINIT`, and pull request 2034.
The tethers	Built 20 October 2026 from the same `tfr9` commit with Go 1.25, without cgo, so each is one file that needs nothing else.

A.3 How It Was Tested

- The TFR/911H firmware is the same program, built from the same source, as the firmware the board these pages were written with has run since 17 October 2026: the two differ only in the date and time of the build that they carry. With it, NitrOS-9 was booted from the FujiNet with **l2fn**; **fnweather**, **fnnews**, **fnstocks**, and **fnbounce** were run, and two full games of **fnbattleship** played.
- The FujiNet firmware was built from fujinet-firmware commit `dca3ee4a3`, whose source is identical to that of the firmware on the FujiNet those tests used; that firmware was flashed with PlatformIO, not with FujiNet-Flasher from this zip.
- The disks were made by the same Makefile, from the same source, as the disks those tests used. `level2-fn.dsk` has the same `OS9Boot`, byte for byte, and the same 114 commands as the disk those tests booted from the FujiNet, but none of the tester's own files.
- `run-tfr.sh` was checked on macOS, with a stand-in for the tether. The tethers were built from the same source as the tether that ran the tests, but these builds have not yet been run with the board.

A.4 Source and Licenses

TFR/9	`github.com/dillera/tfr9`, branch `nitros9-l2-tfr911`; upstream, `github.com/strickyak/tfr9`. MIT License (`licenses/LICENSE-tfr9-MIT.txt`); the FujiNet programs and fujinet-lib for NitrOS-9, in `v4/nitros9-l2/fujinet`, GNU General Public License, version 3 (`licenses/LICENSE-GPL-3.0.txt`).
NitrOS-9	`github.com/dillera/nitros9`, a fork of the NitrOS-9 Project's source, under the NitrOS-9 Project's terms.
FujiNet firmware	`github.com/FujiNetWIFI/fujinet-firmware`, commit `dca3ee4a3`. GNU General Public License, version 3 (`licenses/LICENSE-GPL-3.0.txt`); its source is at that commit.

Building the Release B

Each file of the release can be built again from source. You need what the *TFR Manual and Reference* lists for building NitrOS-9 (**nitros9-l2**(8)): LWTOOLS, ToolShed's **os9**, CMOC, Tcl, and Python; and the Raspberry Pi pico-sdk with the Arm GNU Toolchain for the firmware, Go for the tether, PlatformIO for the FujiNet firmware, and XeLaTeX for the guides. Table B–1 gives the versions this release was built with.

Table B–1 The tools this release was built with

Tool	Version
LWTOOLS (**lwasm**, **lwlink**)	4.22
CMOC	0.1.97
ToolShed (**os9**)	commit `ab6fef5`
pico-sdk	2.3.0
Arm GNU Toolchain	14.2.Rel1
Go	1.25.6
PlatformIO espressif32 platform	6.10.0
XeTeX	TeX Live 2025

B.1 The TFR/911H Firmware and the Disks

```
$ git clone -b nitros9-l2-tfr911 https://github.com/dillera/tfr9
$ git clone https://github.com/dillera/nitros9
$ cd tfr9/v4
$ make -C nitros9-l2 NITROS9_DIR=../../../nitros9 OS9_TOOL=path-to-os9
$ make BUILD-DIR build/tfr911h_v4.uf2
```

The first **make** builds `nitros9-l2/build/tfr911-level2.dsk` (the release's `tether/level2.dsk`), `nitros9-l2/build/tfr911-level2-fn.dsk` (its `sdcard/tfr911/level2-fn.dsk`), and the kernels the firmware holds; the second builds the firmware, `build/tfr911h_v4.uf2`. The firmware's Makefile expects `pico-sdk` beside `tfr9`, and the Arm GNU Toolchain's `bin` directory on your `PATH`.

B.2 The Tethers

```
$ cd tfr9/v4
$ CGO_ENABLED=0 GOOS=darwin GOARCH=arm64 go build -trimpath -ldflags "-s -w" \
      -o tether-darwin-arm64 ./tether/
```

and the same with `GOOS` and `GOARCH` set to `darwin`/`amd64`, `linux`/`amd64`, `linux`/`arm64`, and `linux`/`arm` (with `GOARM=7`).

B.3 The FujiNet Firmware

```
$ git clone https://github.com/FujiNetWIFI/fujinet-firmware
$ cd fujinet-firmware
```

```
$ git checkout dca3ee4a3
$ printf '[fujinet]\nbuild_board = fujinet-coco-devkitc-centipede\n' \
      > platformio.local-rel.ini
$ ./build.sh -y -z -l platformio.local-rel.ini
```

The zip is `firmware/fujinet-COCO-HS-UART-v1.6.2-dev.zip`, with `firmware/release.json`, which is also in the zip.

B.4 The Guides

```
$ make -C tfr9/v4/nitros9-l2 manual install-guide
```

builds `doc/tfr-manual.pdf` and `doc/tfr-install-guide.pdf`. Appendix A's checksums are written into `doc/install.tex` by hand: change them when the files change.

Pins and Settings

C

C.1 Header J3 of the TFR/911H

Table C–1 Header J3

Pin	Marked	RP2350B	Use
1	G8	GPIO 8	Sound: PWM to an amplifier (see the manual's Appendix A)
2	+5V	VBUS	5 V from the USB-C connector
3	G9	GPIO 9	
4	3V3	3V3	3.3 V
5	G10	GPIO 10	
6	GND	GND	Ground: **the FujiNet's ground**
7	G11	GPIO 11	
8	RESET	GPIO 20	The 6309's RESET: connect nothing
9	G12	GPIO 12	UART0 TX: **to the FujiNet's IO33**
10	NMI	GPIO 21	The 6309's NMI: connect nothing
11	G13	GPIO 13	UART0 RX: **from the FujiNet's IO21**
12	HALT	GPIO 24	The 6309's HALT: connect nothing
13	G14	GPIO 14	
14	FIRQ	GPIO 23	The 6309's FIRQ: connect nothing
15	G15	GPIO 15	
16			Not connected
17	G16	GPIO 16	
18	G19	GPIO 19	
19	G17	GPIO 17	
20	G18	GPIO 18	

C.2 The FujiNet's Pins

Table C–2 The DevKitC pins the FujiNet firmware uses

Pin	Header	Use
IO33	J2-8	UART1 RX: DriveWire from the TFR/911H, 921,600 baud
IO21	J3-6	UART1 TX: DriveWire to the TFR/911H
IO5, IO18, IO23, IO19	J3-10, 9, 2, 8	The SD card: CS, SCK, MOSI, MISO
IO15	J3-16	SD card detect (optional)
IO4	J3-13	Bus activity LED (optional)
IO0	J3-14	Button A: the DevKitC's BOOT button
IO1, IO3	J3-4, 5	UART0: the USB bridge, for flashing and the log, 460,800 baud

C.3 The Settings in fnconfig.ini

Table C–3 The settings in the release's fnconfig.ini

Section	Setting	Meaning
`[General]`	`devicename`	The FujiNet's name on your network
	`configenabled`	1: the FujiNet puts its CONFIG disk in drive 0 at power-up, which **l2fn** replaces with device slot 1
	`encrypt_passphrase`	0: the WiFi password is stored as typed
`[WiFi]`	`enabled`	1 to use the WiFi, 0 not to
	`SSID`	Your network's name
	`passphrase`	Its password
`[Network]`	`sntpserver`	Where the FujiNet gets the time
`[Host`*n*`]`	`type`	`SD`, the SD card, or `TNFS`, a TNFS server
	`name`	The server's name (for `SD`, just `SD`)
`[Mount`*n*`]`	`hostslot`	The host slot the disk is on, from 1
	`path`	The disk's path on that host
	`mode`	`r`, read-only, or `w`, read and write

`[Mount1]` is device slot 1, DriveWire drive 0, `/X0`; `[Mount2]` through `[Mount4]` are `/X1` through `/X3`. The FujiNet writes every setting it has when it saves the file; these are the ones this release needs.